

Low-dimensional feature fusion strategy for overlapping neuron spike sorting

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ABSTRACT

Complex action potentials of the brain neurons from extracellular recordings are usually represented as overlapping superimposed spikes resulting from the neuron spike bursts. In this paper, a novel feature fusion strategy is proposed for the high-precision classification of the neuron spikes. Based on the wavelet coefficients and the principal component analysis (PCA) features of the spikes, the eigenvectors of the adjacency matrices are constructed using the Locality Preserving Projections (LPP) algorithm. The fusion adjacency matrix of different features is constructed using different weighted adjacency matrices method. The feature fusion data based on the wavelet coefficients and PCA features are projected from high-dimensional space to low-dimensional space. The experiment results show that the noise level is 0.4, and the accuracy of this method is increased by 9% on average compared with non-fusion feature classification. According to the experimental results, a high-precision classification is obtained based on the feature fusion method with a low-dimensional feature space. The proposed feature fusion method effectively reduces the dimension number of the features and remedies the missing information.

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1. Introduction

Neuroscientists have extensively studied neuron spikes recorded extracellularly [1]. Spike classification poses a great technical challenge for research in neural information science [2]. Neural spike sorting is already used in brain science research. Neural spikes are a comprehensive reflection of the electrophysiological activity of brain cells. The recording, detection, and classification of neural spikes are a hotspot in the field of brain science engineering [3,4]. The extracellular electrode technology is a common method of neural spike recording. However, the simultaneous discharges of multiple adjacent neurons near the microelectrodes often result in overlapping spikes. Therefore, it is essential to separate the spikes according to each neuron.

Related studies have been conducted on the classification of overlapping spikes. The template-matching method has been used to solve the problem of overlapped spikes [1] by using PCA to extract the spike features, which are compared with the template to identify the types of spikes. However, the applicability of this method is limited because of the need for a relevant template. Another approach achieves overlapping spike sorting using neural networks to learn more about the general decision boundaries [2].

This approach is that the neural network must be trained using the spikes data recorded.

The noise-assisted strategy has been used to identify and classify overlapping spikes, and the further classification is performed by a trained learning machine [3]. The independent component analysis (ICA) method can also separate overlapped spikes, but it must meet the requirement that there are more channels of overlapping spikes than the number of spike components [4,7]. The relaxation algorithm has been used to solve the problem of overlapping spikes by using the Fast Fourier Transformation (FFT) to extract the spike features and then matching and identifying the types of spikes [5]. However, this method is sensitive to noise. A single feature method is usually used to identify the spikes. The type of overlapping spike also increases with its number. Therefore, a single feature of spike is often insufficient to characterize the differences between the different types of spikes.

In this paper, a feature fusion strategy is proposed to classify overlapping spikes. This strategy aims at utilizing the advantages of different features to work out different spike extraction methods. It can make up for the deficiency of using a single feature and provide a new measurement mode of the different spikes. Both the wavelet coefficients and the PCA features extraction are widely used in the classification of the spikes. The wavelet coefficients' reflect local difference of the spikes. On the other hand, the PCA features extraction reflects the spike amplitudes. The purpose of the feature fusion is to fully utilize different features of the

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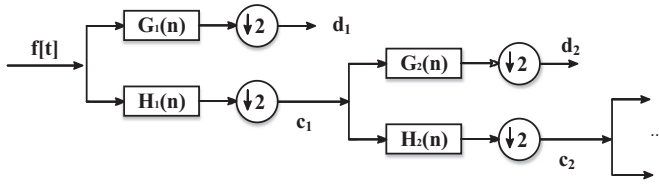


Fig. 1. Schematic diagram of discrete wavelet decomposition, G is the high pass filter coefficient, H is the low pass filter coefficient, $\downarrow 2$ denotes subsampling.

same objects. Therefore, the proposed methods with two kinds of features simultaneously are integrated to construct a new feature vectors using the feature fusion algorithm based on a locality preserving projections (LPP). Finally, the support vector machines (SVM) [6] and K-means method be used to classify the spikes.

2. Representation of spikes with multi-features

2.1. Feature extraction of wavelet coefficients

Wavelet transforms can be applied to any finite-energy function to map the function from the time domain into a compound time–frequency space. The neural spike is decomposed into an orthogonal function space with the wavelet transform, which consist of a scaling function and a wavelet function. The discrete wavelet transform (DWT) is a linear transformation that operates on a data vector whose length is an integer power of 2, transforming it into a numerically different vector of the same length, as shown in Fig. 1. The discrete wavelet decomposition of the low-frequency part can be continued to decompose into a lower-frequency part and a relatively high-frequency part according to the needs of the analysis [9,10].

For any spike, the wavelet decomposition is

$$f(t) = \sum_k c_{j,k} \phi_{j,k} + \sum_k d_{j,k} \psi_{j,k} \quad (1)$$

where $c_{j,k}$ is the k th lower frequency portion of scale j , $d_{j,k}$ is the k th higher frequency portion of scale j , $\phi_{j,k}$ is the scaling function and $\psi_{j,k}$ is the wavelet function, which can be expressed as

$$\begin{cases} \psi_{j,k}(t) = 2^{-\frac{j}{2}} \psi(2^{-j}t - k) \\ \phi_{j,k}(t) = 2^{-\frac{j}{2}} \phi(2^{-j}t - k) \end{cases} \quad (2)$$

where t is the sampling point of the spike, j is the scale parameter and k is the shift parameter. The coefficients of the scaling function and the wavelet function have the relationships, where, c is the lower frequency portion and d is higher frequency portion.

$$\begin{cases} c_{j-1,k} = \sum_{l \in \mathbb{Z}} h(l-2k) c_{j,k} \\ d_{j-1,k} = \sum_{l \in \mathbb{Z}} g(l-2k) d_{j,k} \end{cases} \quad (3)$$

This relationship indicates that both the wavelet function coefficients and scaling function coefficients under scale $j-1$ can be obtained through function coefficients under scale j . $h(l-2k)$ represents a low-pass filter, and $g(l-2k)$ represents a high-pass filter, l is the number of sampling points of spike [6].

Each spike waveform consists of 64 sampling points. As mentioned above, each spike waveform is decomposed into 6 scales using Symlet8 wavelets [8]. Symlet8 wavelets are chosen due to their compact support and orthogonality. Then, the distinguishing features of the spikes are expressed using a few wavelet coefficients. The wavelet coefficients represented with $[c_6, d_6, \dots, d_2, d_1]$ can be obtained with six-level decomposition. c_6 is the approximation level, and $d_6, \dots, d_2, d_1$ are the detail levels.

2.2. Feature extraction of PCA

Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables. The PCA method transforms a set of high dimensional vectors into a lower dimension space. A feature extraction method is used to determine an appropriate subspace mapping in the original feature space.

PCA features extraction is to characterize the maximum change direction of data sets using the ordered orthogonal basic vector set. Then, the original high-dimensional relativity vector can be mapped into a low-dimensional vector space by an eigenvector matrix. The new vector not only reflects the original information but also realize the irrelevance between new indicators [11].

The spikes are defined as $[X_1, X_2, \dots, X_m]$, and each spike consists of an n -dimensional vector that can be expressed as $[x_{i1}, x_{i2}, \dots, x_{in}]$. Then, the whole data set of spikes can be expressed as

$$\mathbf{X} = [X_1 \ X_2 \ \dots \ X_m]^T = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \quad (4)$$

where m, n is number of spikes, the features number for each spike, respectively. The extracted PCA features $\mathbf{Y}_p$ can be expressed with the whole dataset of spikes as:

$$\mathbf{Y}_p = \mathbf{X} \mathbf{G} \quad (5)$$

where matrix $\mathbf{G}$ is the linear mapped matrix, which is composed of the eigenvectors with the maximum eigenvalues of the $\mathbf{X}^T \mathbf{X}$ covariance matrix. The mapped vector of PCA is $\mathbf{Y}_p = [Y_{p1}, Y_{p2}, \dots, Y_{pn}]$, where Y_{p1} represents the indicator composed of the first linear combination of the original variables, that is $Y_{p1} = a_{11}X_1 + a_{12}X_2 + \dots + a_{1m}X_m$. The information quantity extracted from each main component can be indicated by the variance. Y_{p1} is the first principal component, and the mapped vector $\mathbf{Y}_p$ can be defined as:

$$\mathbf{Y}_p = \begin{bmatrix} Y_{p1} \\ Y_{p2} \\ \vdots \\ Y_{pm} \end{bmatrix} = \begin{bmatrix} a_{11}x_{11} + a_{12}x_{12} + \dots + a_{1n}x_{1n} \\ a_{21}x_{21} + a_{22}x_{22} + \dots + a_{2n}x_{2n} \\ \vdots \\ a_{m1}x_{m1} + a_{m2}x_{m2} + \dots + a_{mn}x_{mn} \end{bmatrix} \quad (6)$$

According to the analysis above, we can derive that Y_{pi} and Y_{pj} are uncorrelated, that is, $\text{Cov}(Y_{pi}, Y_{pj}) = 0$. Y_{p1} has the greatest variance among all the linear combinations of $X_1, X_2, \dots, X_m$ and is called the first principal component. $Y_{p2}, Y_{p3}, \dots, Y_{pm}$ are the 2th, 3th, ..., m th principal components, respectively [12].

The PCA features of the spikes are extracted, and a 32-dimensional PCA eigenvector is constructed by the first 32 dimensions of the main component. In the third step of the feature extraction, we obtain the eigenvalues of the sample data covariance matrix. In this paper, the first 32 eigenvalues represent more than 96% of the sum of all eigenvalues. Thus, that PCA eigenvector can characterize more than 96% of the amplitude information of the spikes [13].

3. Feature fusion

3.1. Multiple features

The high-frequency coefficients of the wavelet decomposition indicate the sharp information of the spikes. For the feature extraction with the PCA method, the amplitude of the spike is mapped into a new group of coordinates, resulting in an eigenvector that reflects the amplitude information.

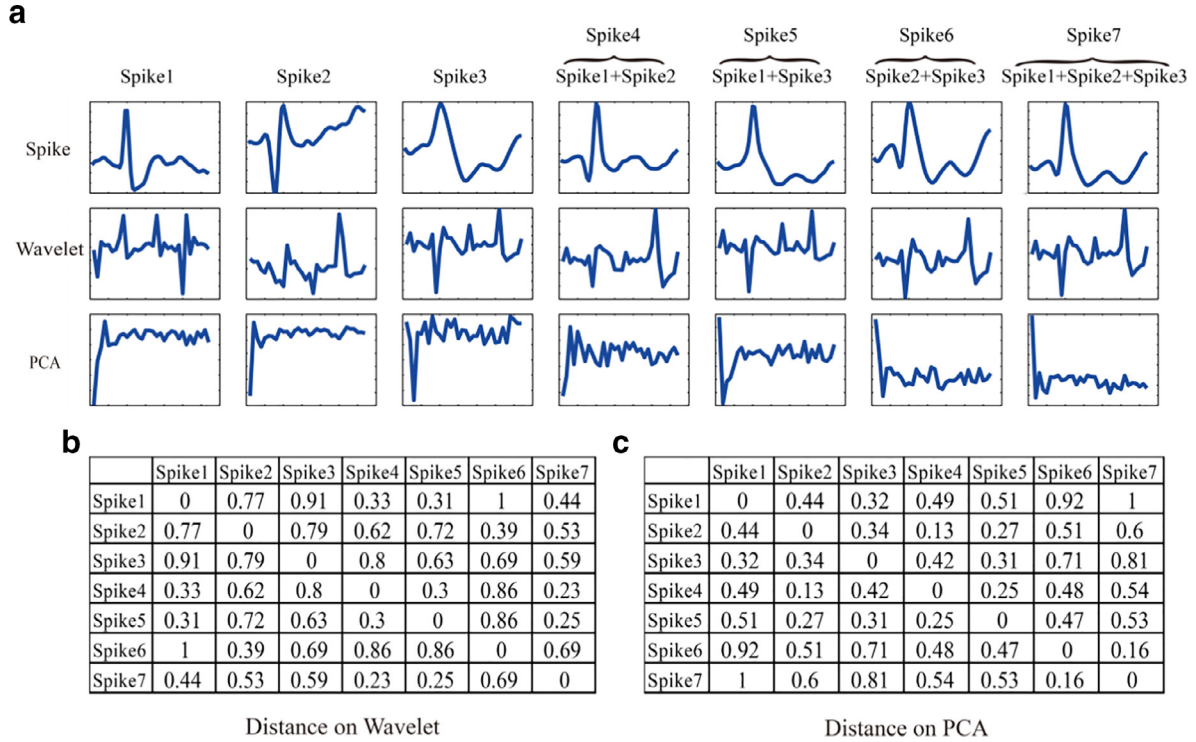


Fig. 2. Complementary characteristics of multiple features of spikes: (a) multiple feature representation of spikes; (b) wavelet coefficient distance measurement of different spikes; (c) PCA features distance measurement of different spikes.

The spike information can be decomposed into the product part of the low-frequency wavelet coefficients $c_{j,k}$ and scaling functions $\phi_{j,k}$ and the product part of the high-frequency coefficients $d_{j,k}$ and wavelet functions $\psi_{j,k}$. The high-frequency wavelet coefficients illustrate the local signal mutation of dissimilar spike shapes, and the PCA features illustrate the amplitude information of dissimilar spike [14]. The fusion information of the sharp and the amplitude of the spike are used to sort spikes. To improve the classification accuracy, the two type features about wavelet and PCA are fused. An additional feature can be a good supplement when one feature is insufficient to represent the difference between spikes. When there is little difference between the sharp information of two different spikes, the wavelet coefficients become insufficient to distinguish between them. Then, we can take advantage of their different amplitudes. On the other hand, there is little difference between the overall profiles of two different spikes, the use of the PCA features may fail. Then, we can compare the sharp information of the two spikes to distinguish them [15]. The fusion feature advantages of wavelet coefficients and PCA method can be achieved.

The distance between the feature vectors of two different spikes is expressed by the 2-norm as follows, P and Q are the feature vectors of the two spikes.

$$t = \|P - Q\|_2 = (|p_1 - q_1|^2 + |p_2 - q_2|^2 + \dots + |p_n - q_n|^2)^{1/2} \quad (7)$$

where n is the total number of spike features.

To analyze the distance between the different spikes features more intuitively, the distance is normalized to 0–1. The maximum and minimum normalization method is adopted. The calculation results are shown in Fig. 2. P and Q represent the feature vectors of different spikes. In this paper, we extract both 32 dimensions of wavelet coefficients and 32 dimensions PCA features of spikes. Fig. 2 shows the non-overlapped spikes, double-overlapped spikes, and triple-overlapped spikes with feature fusion. As shown in

Fig. 2(b), the distance between the wavelet coefficients eigenvector of spike 5 and that of spike 7 is 0.25, which indicates a tiny difference between the wavelet coefficient of these two spikes. However, the distance between the PCA features of spike 5 and spike 7 is 0.53 in Fig. 2(c). Therefore, the PCA features have good complementarity with the wavelet coefficients.

3.2. Locality preserving projection

The Locality Preserving Projection (LPP) algorithm [16] is used to realize the feature fusion of the dissimilar feature spikes. The LPP algorithm achieves the linear approximation function based on the non-linear Laplacian eigenmap. It can preserve local neighborhood information through the notion of the Laplacian of the graph. The collection of wavelet coefficients and PCA features is fused using the LPP algorithm. The essence of method proposed is the fusion of some different characteristic to obtain a new characteristic representation and realize the complementation of the different features. Then, a transformation matrix is computed and maps the data points to a subspace (see Fig. 3).

3.3. Feature fusion based on LPP

The feature fusion based on LPP is used for unsupervised classification [17]. The feature fusion algorithm based on locality preserving projections construct a local neighborhood structure for the characteristics of each feature. Based on the local neighborhood structure with different features, a global alignment is constructed to encode the objective function for each local neighborhood structure of the feature. An alignment matrix through the whole alignment stage is constructed with all eigenvector global alignments from different local neighborhood structures. To consider the complementarity of multi-features, a weighted method is employed to linearly combine the global alignments for different features to obtain the realized alignment matrix. Then, the low-dimensional fea-

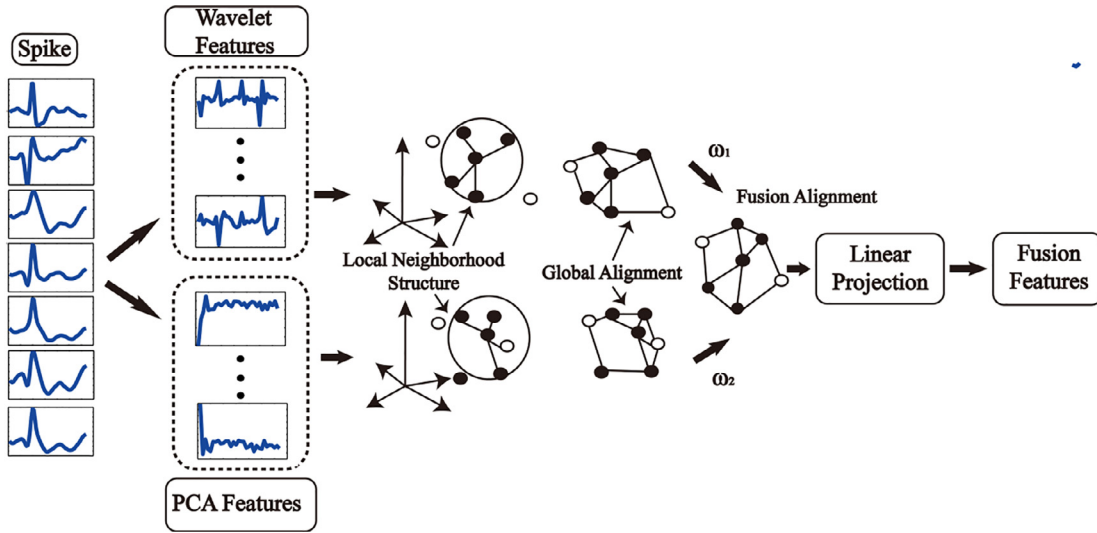


Fig. 3. Process of the feature fusion based on the wavelet coefficients and PCA features.

ture fusion space is obtained. Finally, the spike feature can be realized by measuring the dissimilarity in this subspace of the feature.

Based on the wavelet coefficients and PCA features of the spikes extracted, the adjacency graphs can be constructed from the different features. Adjacent matrix $\mathbf{W}$ represents the adjacency graphs, and it is a measurement of the similarity between different spikes. For different features, the similarity measurement matrix is different, and its function of the classification of spikes is also different. Therefore, these weights are introduced to reflect the importance of different features. A weighted strategy of feature fusion is derived to reflect the relative importance of different features. The fusion matrix is obtained by the fusion of attributed weights. Then, the fusion feature is obtained by linear mapping [18]. This process can be expressed as,

$$\bar{\mathbf{W}} = \sum_{k=1}^K \omega_k \mathbf{W}^k, 0 < \omega_k < 1 \text{ and } \sum_{k=1}^K \omega_k = 1 \quad (8)$$

where k is the types of spikes' feature sets. ω_k is the weights of different features.

The optimization problem is a complicated problem related to matrix $\mathbf{A}$ and weights ω .

$$\min \delta(\mathbf{A}, \omega) = \text{tr} \left(\left[\sum_{i,j} \|\mathbf{A}^T \mathbf{x}_i - \mathbf{A}^T \mathbf{x}_j\|^2 \sum_{k=1}^K \omega_k \mathbf{W}_{ij}^k \right] \right) \quad (9)$$

The optimization problem is solved by the following method. We obtain the following optimization problem by fixing ω . This optimization problem can be solved by a method solving eigenvalue showed in following paper.

$$\min \delta(\mathbf{A}) = \text{tr} \left(\left[\sum_{i,j} \|\mathbf{A}^T \mathbf{x}_i - \mathbf{A}^T \mathbf{x}_j\|^2 \sum_{k=1}^K \omega_k \mathbf{W}_{ij}^k \right] \right) \quad (10)$$

Similarly, we can get following optimization problem by fixing $\mathbf{A}$. It can be solved by a quadric programming method.

$$\min \delta(\omega) = \text{tr} \left(\left[\sum_{i,j} \|\mathbf{A}^T \mathbf{x}_i - \mathbf{A}^T \mathbf{x}_j\|^2 \sum_{k=1}^K \omega_k \mathbf{W}_{ij}^k \right] \right) \quad (11)$$

$$0 < \omega_k < 1 \quad (12)$$

$$\sum_{k=1}^K \omega_k = 1 \quad (13)$$

Given the tolerance error $\delta_{\mathbf{A}}$, δ_{ω} , we can iterative compute (10) and (11) until satisfy the following formula.

$$|\delta_{\mathbf{A}}^i - \delta_{\mathbf{A}}^{i-1}| \leq \delta_{\mathbf{A}} \text{ and } |\delta_{\omega}^i - \delta_{\omega}^{i-1}| \leq \delta_{\omega} \quad (14)$$

So, we can obtain the weights ω though the above method [18]. Next, we will discuss how to solve the matrix $\mathbf{A}$ in case of fixing ω .

If we fix the parameter ω , the problem in (9) is an eigenvalue one. Then, the new objective function is minimized,

$$\mathbf{A} = \arg \min \left[\sum_{i,j} \|y_i - y_j\|^2 \sum_{k=1}^K \omega_k \mathbf{W}_{ij}^k \right] \quad (15)$$

y_i is the fusion feature mapped by matrix $\mathbf{A}$. The objective function minimized can ensure that the adjacency structure of the feature space before and after mapping does not change, that is, $\mathbf{Y}^T = \mathbf{A} \mathbf{X}^T$. y_i is the first column of $\mathbf{Y}$, and x_i is the first column of the spikes feature sets $\mathbf{X}$. Therefore, $\mathbf{A}$ is furtherly transformed:

$$\begin{aligned} \mathbf{A} &= \arg \min \left[\sum_{i,j} \|\mathbf{A}^T \mathbf{x}_i - \mathbf{A}^T \mathbf{x}_j\|^2 \sum_{k=1}^K \omega_k \mathbf{W}_{ij}^k \right] \\ &= \arg \min \left[\sum_{k=1}^K \sum_i \mathbf{A}^T \mathbf{x}_i \omega_k D_{ii}^k \mathbf{x}_i^T \mathbf{A} - \sum_{k=1}^K \sum_i \mathbf{A}^T \mathbf{x}_i \omega_k W_{ij}^k \mathbf{x}_j^T \mathbf{A} \right] \\ &= \arg \min [\mathbf{A}^T \mathbf{X} (\bar{\mathbf{D}} - \bar{\mathbf{W}}) \mathbf{X}^T \mathbf{A}] \\ &= \arg \min [\mathbf{A}^T \bar{\mathbf{L}} \mathbf{X}^T \mathbf{A}] \end{aligned} \quad (16)$$

In Eq. (10), $\bar{\mathbf{L}}$ is a Laplace matrix, $\bar{\mathbf{L}} = \bar{\mathbf{D}} - \sum_{k=1}^K \omega_k \mathbf{W}^k$. $\bar{\mathbf{D}}$ is a diagonal matrix, and $\bar{D}_{ii} = \sum_{j,k=1}^K \omega_k W_{ij}^k$. When $\mathbf{A}^T \bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} = 1$, this problem can be solved by a Lagrange multiplier as follows:

$$\mathcal{L}(\mathbf{A}, \lambda) = \mathbf{A}^T \bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} - \lambda (\mathbf{A}^T \bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} - 1) \quad (17)$$

where λ is the eigenvalues of generalized eigenvector problem: $\bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} = \lambda \bar{\mathbf{D}} \mathbf{X}^T \mathbf{A}$. Take the derivative,

$$\partial \mathcal{L}(\mathbf{A}, \lambda) / \partial \mathbf{A} = 2 \bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} - 2 \lambda \bar{\mathbf{D}} \mathbf{X}^T \mathbf{A} = 0$$

Then,

$$\bar{\mathbf{L}} \mathbf{X}^T \mathbf{A} = \lambda \bar{\mathbf{D}} \mathbf{X}^T \mathbf{A} \quad (18)$$

Eigenvalues $\lambda_1 < \lambda_2 < \dots < \lambda_n$ can be obtained with solution of eigenvalues, and mapping matrix $\mathbf{A}$ consists of eigenvectors a_i

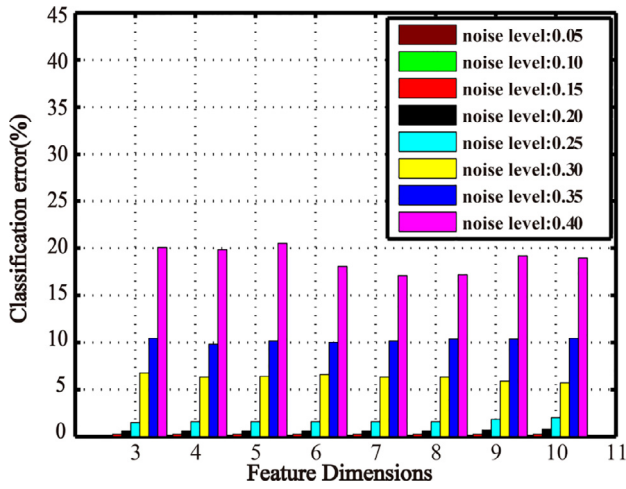


Fig. 4. Classification performance of overlapping example 1 data with the different feature dimensions.

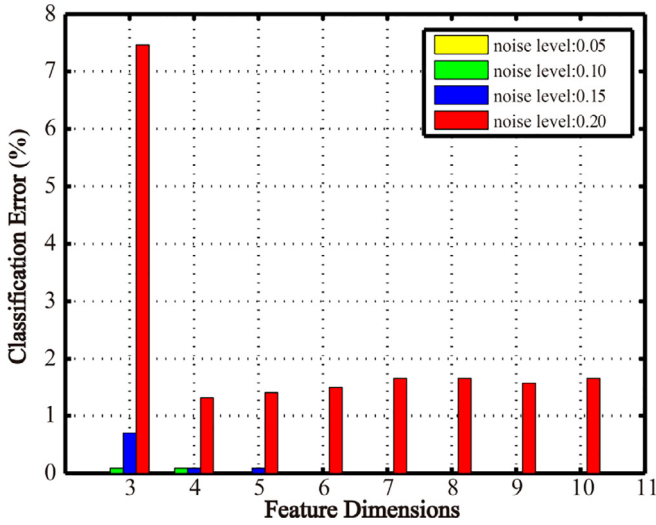


Fig. 5. Classification performance of overlapping example 3 data with the different feature dimensions.

($i = 1, 2, \dots, n$) corresponding to eigenvalues. From expression (12), mapping matrix $\mathbf{A}$ preserves the local neighborhood structure of the wavelet coefficients and the PCA features, which are characterized by mixed adjacent matrix $\mathbf{W} = \omega_1 \mathbf{W}_1 + \omega_2 \mathbf{W}_2$. Using the linear transform $\mathbf{Y}_f^T = \mathbf{A} \mathbf{X}_f^T$, we can obtain the fusion eigenvector data set $\mathbf{Y}_f$. The wavelet coefficients and PCA features can be fused by the mapping matrix $\mathbf{A}$, which not only realizes the complementation of different features, but also achieves dimensional reduction. The fusion feature can be obtained by the expression $\mathbf{Y}_f^T = \mathbf{A}(\mathbf{X}_w^T \mathbf{X}_p^T)^T$, where $\mathbf{X}_w$ is the wavelet coefficients set, and $\mathbf{X}_p$ is the PCA features set.

4. Experimental results

4.1. Experimental data

In this paper, the experimental spike data set includes the artificially synthesized dataset and the recording spike dataset [19,20]. The artificially synthesized dataset includes three types of basic spikes under different noise levels. These overlapped spikes are formed by artificially superposed with two basic spikes or three basic spikes. 360 waveforms per basic spike were generated, 120 waveforms per double overlap were generated, and 120 waveforms

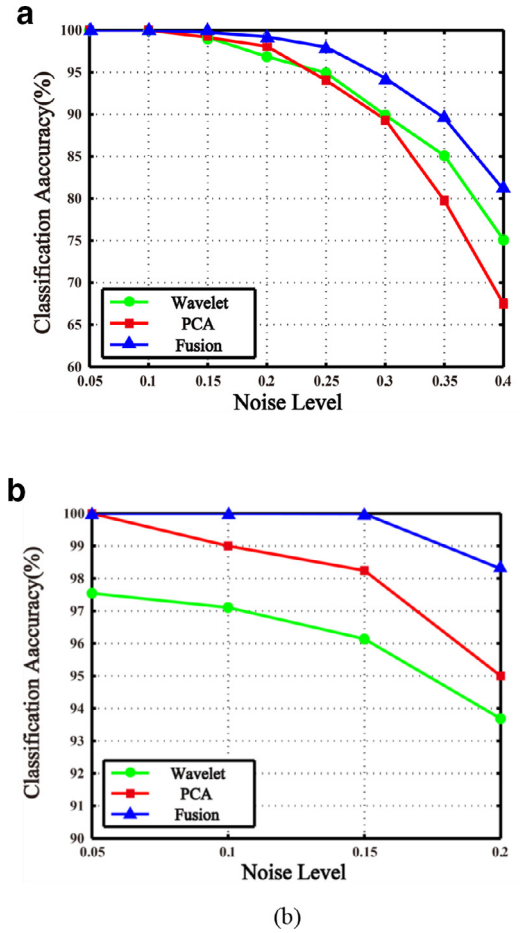


Fig. 6. Comparison of classification accuracy of the wavelet coefficients, PCA features and the fusion features for (a) example 1 and (b) example 3.

of per triple overlaps were generated. The support vector machine is trained for classification. The training dataset is four types, example 1, example 2, example 3, and example 4, each with 60 spikes. The recording spike dataset is a continuous time series of spikes that includes four overlapping datasets with different classification difficulty degrees. Both the support vector machine and K-means are adopted as classifiers in the experiment [21].

4.2. Experimental results and discussion

The process of the experiment in this paper is as follows. First, each spike waveform is decomposed into 6 scales using the Symlet8 wavelet. Next, we choose 32 dimensions of wavelet coefficients from the detail levels. Then, we extract PCA features from spikes and choose the first 32 dimensions of the principal components from the PCA features, which characterize more than 96% of the spikes' amplitude information. Finally, the wavelet coefficients and PCA features are fused into mixed features based on locality preserving projections [22].

4.2.1. Results and discussion of synthetic dataset

Wavelet coefficients and PCA features are two popular techniques used in extracting discriminative spike features, but they both have limitations in characterizing spikes. A fusion method is proposed in which the wavelet coefficients and PCA features are fused to achieve the complementation of advantages to overcome this limitation. The different features have different feature spaces. Therefore, a fusion algorithm is needed to express different features in the same feature space.

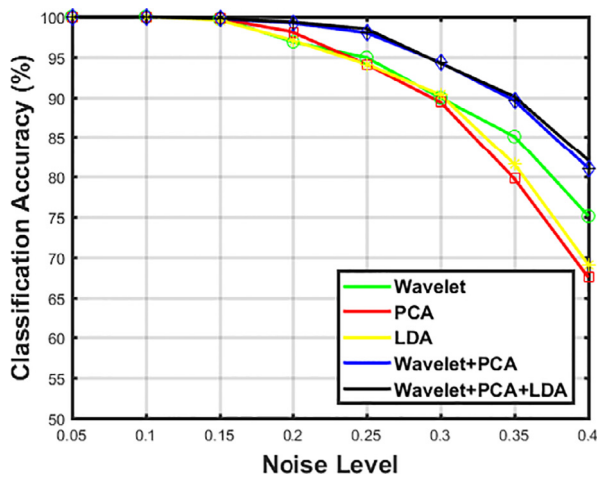


Fig. 7. Comparison of classification accuracy of the fusion features based on the wavelet coefficients, PCA, and LDA for dataset example 1.

The dimensionality of features is an important evaluation index in classification, so it is meaningful to achieve good performance using low dimensions of features. Those overlapped spikes example 1 are comparatively easy to classify. The classification accuracy under different noise levels and different feature dimensions are shown in Fig. 4. When the feature dimension is less than or equal to two, the error rate is too high and neglected. When the feature dimensions are more than three, the accuracy of the fusion features tends to be stable under different noise levels. Therefore, the features of 3 dimensions are sufficient in characterizing differences among dissimilar spikes. For those overlapped spikes consisting of example 3, which are comparatively difficult to classify, their classification accuracy under different noise levels and different feature dimensions is shown in Fig. 5. When the feature dimensions are greater than 3, the classification accuracy is greater than 98%.

Fig. 6 shows the classification accuracy of single features and fusion features. When the noise level is lower than 0.2 in Fig. 6(a), the accuracy of both the single features and the fusion features is more than 95%. As the noise level increases, the classification accuracy is seriously affected by noise, which increases the difficulty of the classification. When the noise level is over 0.25, the classification accuracy using the wavelet coefficients and PCA features

rapidly decrease as shown in Fig. 6(a). On the other hand, the feature fusion method maintains a high accuracy of classification until the noise level is 0.2 compared to the other two methods in Fig. 6(b). Therefore, the feature fusion method with the wavelet coefficients and PCA features realize high classification accuracy based on the LPP algorithm. The experimental results show that the classification accuracy of the fusion features at different noise levels remains more than 80% for example 1 and 98% for example 3.

Fig. 7 shows the classification accuracy of single features and fusion features. When the noise level is lower than 0.2, the accuracy of both the single features and the fusion features is more than 95%. As the noise level increases, the classification accuracy is seriously affected by noise, which increases the difficulty of the classification. When the noise level is over 0.25, the classification accuracy using the wavelet coefficients, PCA and LDA feature rapidly decrease as shown in Fig. 7. On the other hand, whether the fusion feature based on wavelet, PCA and LDA or fusion features based on wavelet, PCA maintains a high accuracy of classification until the noise level is 0.4 than the other three non-fusion methods. Three kinds of feature fusion based on wavelet, PCA and LDA have very few improving than two kinds of feature fusion based on wavelet and PCA.

4.2.2. Results and discussion of recording spike data set

The real datasets of the neural spike are employed to compare the classification of the wavelet coefficients, PCA features, and fusion features. In this experiment, the feature dimension is ten dimensions, and the classifier is the support vector machine (SVM) and *K*-means. Those continuous time series datasets are the spike data with different noise levels. Table 1 shows the experimental results without overlapped spikes, and Table 2 shows the experimental results with overlapped spikes.

Considering the overlapping spike in Table 2, the fusion features have a more satisfactory performance compared with single features, regardless of whether a support vector machine or *K*-means used. The advantage of fusion features becomes more obvious with the rise in the noise level. Among the dataset for the low noise level of 0.40, the accuracy of the fusion features using *K*-means reaches 74.6308% in Table 2, which is superior to the 68.6946% achieved by wavelet coefficients and the 62.7584% achieved by PCA features.

In some cases, the classification accuracy of the PCA features may be higher than that of the wavelet coefficients. For instance, in

Table 1
Comparison of classification performance with non-overlapping data sets.

Example	Noise level	Spike number	SVM			K-means		
			Wave	PCA	Fusion	Wave	PCA	Fusion
Example 1	005	2729	100	100	100	100	100	100
	010	2753	100	100	100	100	100	100
	015	2693	99.8515	99.7772	99.8515	99.8143	85.0353	99.8143
	020	2678	97.7595	99.1785	99.8506	97.8715	98.4690	98.0583
	025	2586	94.8956	94.6249	98.1825	94.1222	95.4370	94.8569
	030	2629	90.5287	92.0882	93.3435	87.2575	87.6379	90.1864
	035	2702	83.1236	82.9756	87.1207	78.5344	73.7232	86.4175
	040	2645	76.7864	74.4802	81.0586	72.1739	63.8563	80.4915
Example 2	005	2619	100	100	100	100	100	100
	010	2694	96.8448	98.5152	99.9258	95.1373	97.9955	99.4432
	015	2648	88.2175	91.9562	98.3761	77.4547	91.8051	93.3535
	020	2715	80.442	84.9724	94.1805	56.2431	78.1584	81.8048
Example 3	005	2616	99.0443	100	100	98.9679	100	100
	010	2638	91.8878	98.2563	97.953	88.5898	96.3988	97.7255
	015	2660	82.0677	86.7293	92.406	55.2256	80.3008	87.8195
	020	2624	72.2561	77.439	82.0884	47.2561	59.8323	69.5122
Example 4	005	2535	97.8698	99.6844	100	92.3866	98.1460	99.9606
	010	2742	92.0131	93.7637	99.1612	73.3771	83.9168	84.4274
	015	2631	86.8111	82.7822	86.165	79.1334	71.9878	76.0927
	020	2716	78.1664	73.3063	76.3623	63.7703	61.7084	65.0221

Table 2
Comparison of classification performance based on overlapping data sets.

Example	Noise level	Spike number (overlapping)	SVM			K-means		
			Wave	PCA	Fusion	Wave	PCA	Fusion
Example 1	005	3514(785)	96.7843	98.5202	98.6625	96.3574	98.6910	98.7479
	010	3522(769)	98.0125	98.6939	98.9779	96.6780	92.4759	98.8927
	015	3477(784)	96.52	97.0664	98.2169	95.9448	98.5620	98.2744
	020	3474(796)	94.8187	95.5959	97.3518	94.7035	90.9614	96.8336
	025	3298(712)	92.6319	91.9951	95.4518	91.4190	95.3911	92.9048
	030	3475(846)	87.1367	88.2014	88.6619	83.6547	90.5899	88.6331
	035	3534(832)	79.8529	80.4754	83.2201	75.2405	75.2688	83.3899
	040	3386(741)	73.922	72.593	78.1158	68.6946	62.7584	74.6308
Example 2	005	3410(791)	92.4633	93.871	92.2287	87.8299	86.1584	93.0205
	010	3520(826)	81.8891	91.3636	92.9261	82.8977	85.0568	88.6080
	015	3411(763)	81.0026	85.2829	89.7977	80.0352	83.2600	90.7945
	020	3526(811)	74.135	80.1475	86.245	72.1214	67.7255	75.3261
Example 3	005	3383(767)	95.1227	98.2855	97.5761	93.7334	96.6302	98.2855
	010	3448(810)	88.1381	96.0557	94.8086	82.4246	94.2865	94.9536
	015	3472(812)	78.341	82.6613	87.8456	52.6498	82.8917	83.9574
	020	3414(790)	69.8301	73.6965	78.5882	44.8448	58.4066	66.1101
Example 4	005	3364(829)	85.1665	87.3662	87.396	85.3746	85.5529	85.6718
	010	3462(720)	89.1103	83.5066	88.0705	87.7816	73.1369	89.7747
	015	3440(809)	75.3488	72.6163	74.6512	61.6860	85.7558	84.9709
	020	3493(777)	69.3673	65.4738	67.2488	57.4578	58.4311	61.4658

Table 2, among the dataset of the low noise level of 0.25, the classification accuracy of the wavelet coefficients is 92.6319%, higher than the 91.9951% achieved by the PCA features. On the other hand, the classification accuracy of the wavelet coefficients can also be higher than that of the PCA features in some cases. For instance, in Table 2, among the dataset of the low noise level of 0.2, the classification accuracy of the PCA features is 95.5959%, higher than the 94.8187% achieved by wavelet coefficients. Therefore, each of those two features has its own advantage for different datasets and different noise levels.

The fusion features combine the advantages of the localized mutation differences among dissimilar spikes characterized with the wavelet coefficients and the global differences among dissimilar spikes characterized with PCA features. The two types of different features each provide a good supplementation to the other's insufficiencies, when one feature lacks the ability to represent the difference between spikes. The feature fusion method realizes higher classification accuracy than that of single features.

With the appearance of the overlapping spikes, the difficulty of characterizing different spikes also increases. To sum up, the fusion features contains more information than single feature and achieves a better and more stable performance.

5. Conclusions

The feature fusion based on the wavelet coefficients and PCA features with the locality preserving projections algorithm is proposed for the overlapping spike sorting. The feature fusion method realizes the overlapping spike classification based on the wavelet coefficients for characterizing localized mutations and PCA for characterizing the global amplitude differences.

The feature fusion data based on the wavelet coefficients and PCA features are projected from high-dimensional space to low-dimensional space for high classification efficiency. This method effectively reduces the number of dimensions of the features and remedies the missing information. The feature fusion method also completes the spike classification to achieve higher accuracy with noisy datasets. Our motivation regarding the strategy of data fusion to extract spike features in overlapping datasets has been verified by the experimental results. The feature fusion strategy achieves the effective classification of different overlapping spikes using the complementary wavelet coefficients and PCA features.

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